Institute of Technology of Cambodia

Département Of Industrial and Mechanical Engineering

Subject: Dynamic of Gas

Topic: Super Sonic Flight

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Academic Year: 2026-2027

ACKNOWLEDGEMENT

We would like to express our deepest gratitude to **M. Seyhak Dara**, lecturer of the ***Dynamics of Gas* course**, for his invaluable guidance, encouragement, explain the process of Supersonic flight and constructive feedback throughout the duration of this project. His expertise in compressible flow and propulsion systems provided us with the theoretical foundation and practical insight necessary to complete our study on supersonic flight and ramjet design.

We also extend our sincere thanks to the **Institute of Technology of Cambodia** and the **Department of Industrial and Mechanical Engineering** for providing the academic environment, resources, and facilities that made this research possible. The support of the institution allowed us to access computational tools such as MATLAB, CAD, and CFD software, which were essential for conducting simulations and validating our results.

Our appreciation goes to our group members for their teamwork, dedication, and collaborative spirit. Each member contributed unique skills, from coding and simulation to design and analysis, ensuring the success of this project.

Finally, we acknowledge the role of engineering software and computational tools MATLAB for numerical simulation, SOLIDWORKS for CAD modeling, and ANSYS/CFD for aerodynamic analysis. These tools allowed us to bridge theory with application, producing results that are both academically rigorous and practically relevant.

.

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Supersonic Flight

I. Introduction

1. Definition

Supersonic flight occurs when an aircraft travels at a speed equal to or greater than Mach 1, where compressibility effects of air become significant. The development of supersonic and high-speed aircraft is motivated by the need to reduce travel time and improve performance in aviation applications. Achieving such speeds, however, introduces aerodynamic challenges that do not appear in subsonic flight.

At supersonic speeds, the airflow experiences rapid changes in pressure, density, and temperature, resulting in shock waves and increased drag. These phenomena strongly influence aircraft design, propulsion efficiency, and structural requirements. Understanding compressible flow behavior is therefore essential for the development of efficient and practical supersonic aircraft.

Figure 1: The sample of supersonic flight

A **supersonic aircraft** is an aircraft capable of flying faster than the speed of sound, known as Mach 1. Supersonic aircraft were developed mainly for military and research purposes, and only a limited number have been used in civil aviation. The most common examples of supersonic aircraft today are military fighter jets.

Mach is the speed at which an object moves **equal to the speed of sound** in the surrounding air. The speed of sound is **not constant**; it mainly depends on air temperature. At sea level (about 15 °C), Mach 1 is approximately **343 m/s** (≈ **1,235 km/h**). At higher altitudes, where the air is colder, Mach 1 is lower.

- **Mach < 1 → Subsonic**
- **Mach = 1 → Sonic**
- **Mach > 1 → Supersonic**

Mach number is important because when an aircraft approaches or exceeds Mach 1, **compressibility effects become dominant**, leading to shock waves, sudden pressure changes, and increased aerodynamic drag.

2. Objective

The goal of this case study is:

- Applying the Gas Dynamic into the real situation
- To understand the basic concept of airplane
- To know how to design and apply theories (Isentropic, shock flow…) into the case study
- Improving the knowledge of material that fit with environment
- To explore about the airfoil, tail airfoil, and ramjet of airplane
- Looking out the Cd, Cl, Mach Number, Velocity, Static pressure, and etc.
- Analyze and discussion about the result

3. Scope of Work

- **Team Responsibilities**

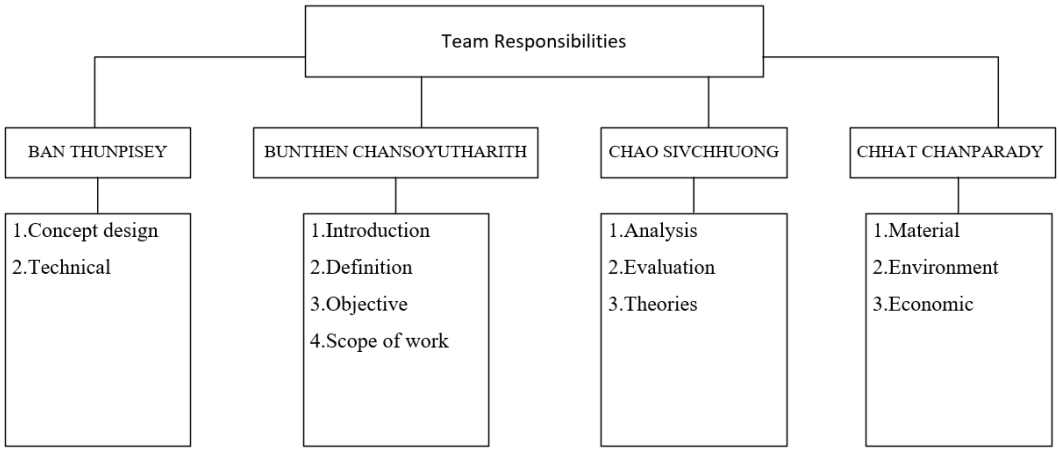

Figure 2: The Responsibilities

- **Flow of Works**

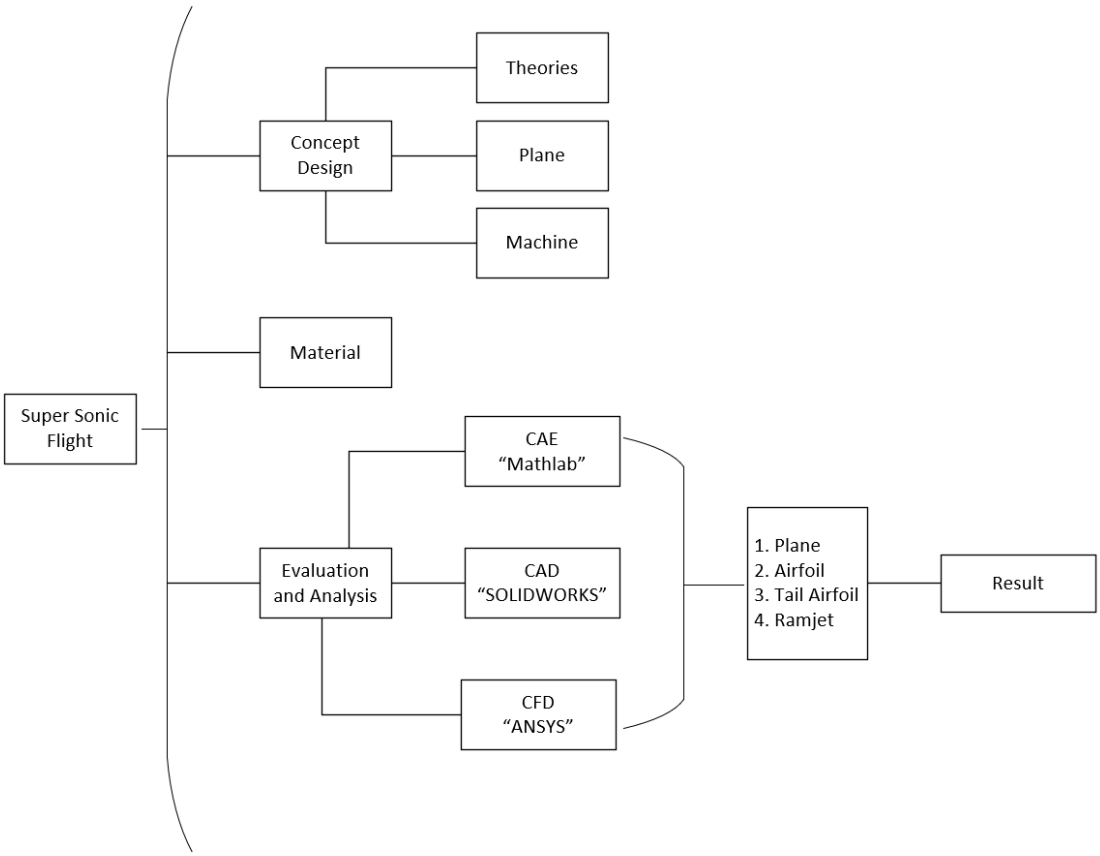

Figure 3: The Flow of Works

II. Review

This project has so far defined supersonic flight, outlined objectives, and begun computational analysis using MATLAB for ramjet flow. Previous studies on Converging–Diverging ramjet have established methods for calculating Mach number distributions and flow properties under supersonic conditions.

- Supersonic flight definition: Clearly explained as speeds equal to or greater than Mach 1, where compressibility effects of air dominate. This sets a strong foundation for understanding the physics involved.
- Motivation for supersonic aircraft: Nicely highlights reduced travel time and improved aviation performance, linking technical goals to practical benefits.
- Aerodynamic challenges: Well, -stated issues like shock waves, drag increase, and rapid changes in pressure, density, and temperature. These are crucial for design considerations and give context for later sections.

- Mach concept: The explanation of Mach number and its dependence on air temperature is accurate and educational. The numerical example at sea level (Mach 1 ≈ 343 m/s) adds clarity.
- Classification summary: The bullet points (Subsonic, Sonic, Supersonic) are concise and effective for quick reference.
- Visuals: The aircraft images reinforce the definition and provide real-world context, though they could be annotated more to highlight features related to supersonic flight.

III. Methodologies

1. Concept Design

1.1. The Flight

The aerodynamics of the F-16 are designed to balance high-speed supersonic flight with extreme agility at lower speeds. The F-16 was the first fighter jet to use a side-mounted control stick. The stick lets pilots rest their arm while flying, giving them better control of the jet in high-G maneuvers. F-16 was the first aircraft designed specifically to prioritize energy-maneuverability theory. This was an application of the principles of thermodynamics to aircraft metric, which uses physics to maximize an aircraft's ability to change altitude and speed while turning. Energy-Maneuverability (E-M) Theory and its application to the F-16's design, there are three key steps:

1. **Energy Height:** It focus on the total energy; this represents the total mechanical energy of the plane (Potential + Kinetic) per unit of weight.

Formula: $he = (h+v^2)/2g$

2. **Specific Power:** It determines if the aircraft can climb or accelerate while performing a specific maneuver.

$\boldsymbol{Ps}$: represent Specific Excess Power.

Formula : Ps = V(T-D) / W

• If Ps >0: The plane can acceleration or climb while maintain its turn

• If Ps =0: The plane is at sustained performance, it can't get faster or higher without stopping the turn.

• If Ps <0: The plane is bleeding energy, it's losing speed or altitude to make a tight maneuver.

3. **Comparative Analysis:** The final step in the design concept is to overlay the F-16's E-M diagram on top of an opponent.

• The "Winning Zone": If the F-16's Ps = 0 line is higher than the enemy's, the F-16 can turn at a rate that the enemy cannot match without slowing down.

• Tactical Result: Designers adjusted the F-16's wing shape and weight until its "Sustained Turn" line Ps = 0 was higher than its competitors in the transonic range (Mach 0.6 to 1.2), where most dogfights occur.

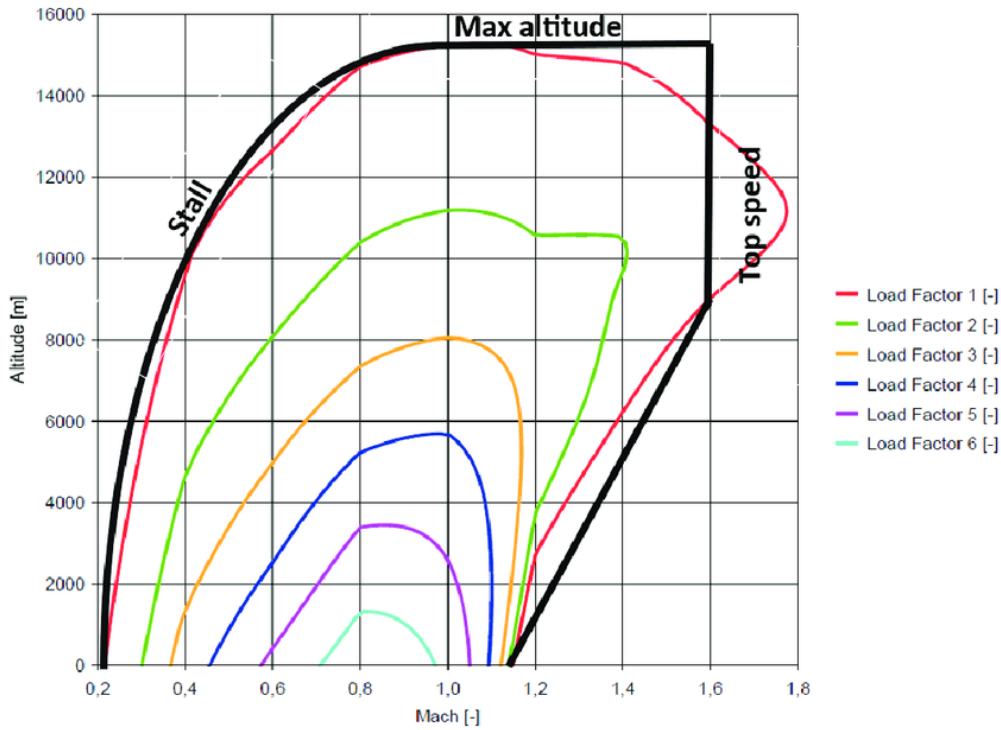

Figure 4: E-M diagram

1.2. The Fuselage

Fuselage (Main body of the aircraft): The F-16's body is not built as one continuous tube but as three separate assemblies bolted together.

1. **Forward fuselage:** assembly housing the aircraft's critical sensor, pilot interface, and primary weapon systems. Key components of the forward fuselage:

- Radom and Radar: A composite cone that is transparent to radio waves, protecting the radar antenna.
- Avionics Bays: Located beneath and behind the pilot, these bays are cooled by a dedicated Environmental Control System (ECS) to prevent sensitive computers from overheating.
- Cockpit & Canopy: Features a single-piece, frameless bubble canopy providing 360° visibility.
- Avionics Bays: Located behind and beneath the cockpit, these bays house the flight control computers (FLCC) and modular mission computers.

2. **Center Fuselage:** the primary load-bearing section of an aircraft. Core function and feature:

- Wing Attachment: Contains the heavy structural members (spars and frames) that securely anchor the wings to the airframe.
- Main Landing Gear: Often provides the storage wells and attachment points for the main

landing gear.

- Load Distribution: It must withstand the highest stresses, including bending from the wings and torsional loads from flight maneuvers.
- Fuel Storage: the blended wing-fuselage design allows for significant internal fuel capacity in this area, which supports its high-g (9g) maneuverability.
- M61A1 Vulcan Cannon: An internal 20mm multi-barrel cannon is mounted on the left side of the forward fuselage.

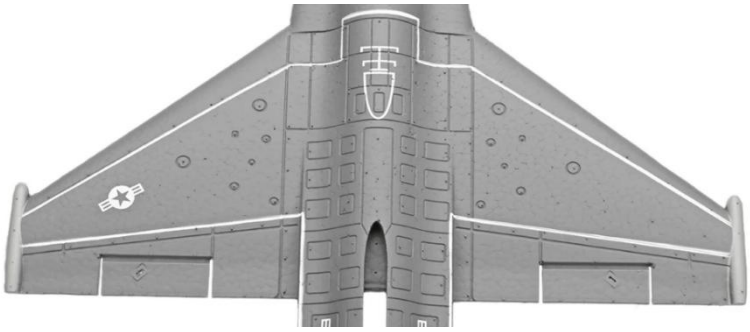

Figure 5: The Fuselage

3. **Aft Fuselage:** Is the rear structural section that supports the engine exhaust, stabilization surfaces, and emergency recovery systems. Key Components & Structural Features:

- Empennage (Tail Assembly): The aft section houses the vertical stabilizer (fin) and all-moving horizontal stabilators.
- Engine Ramjet: modulates to control thrust.
- Speed Brakes: Positioned on either side of the engine ramjet, these two panel split-flap brakes extend outward to rapidly increase drag.
- Emergency Tailhook: A critical safety tool mounted underneath for arrested emergency landings or aborted takeoffs using runway cables.
- Parachute Fairing: is a small fairing at the base of the vertical stabilizer houses a drag chute or electronic countermeasures (ECM) equipment.
- Ventral Fins: Located on the bottom surface rearward of the wings, these two fins provide directional stability during high-angle-of-attack maneuvers.

Figure 6: AFT Fuselage

1.3. Technical

The F-16 is a Mach 2.0+ fighter capable of reaching altitudes over 50,000 feet. With a 1.1 thrust-to-weight ratio, it can accelerate during vertical climbs and is built to sustain extreme 9.0 g maneuvers. Though it weighs only 19,000 lbs empty, its robust design allows for a maximum takeoff weight of 42,300 lbs.

The F-16 uses an advanced digital "nervous system" to collect and process data.
Because the plane is aerodynamically unstable, it requires constant data feedback to remain airborne.

The F-16 has a small tube on its nose called a pitot tube. It works by sensing the pressure of air as the jet moves forward and compares it with the surrounding air pressure. This difference tells the aircraft its speed, which is vital for safe flying.

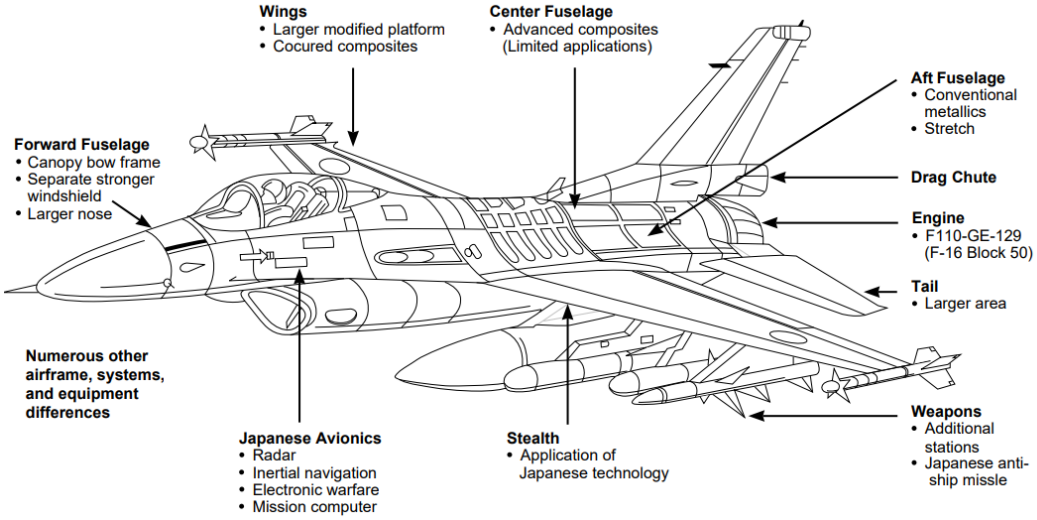

Figure 7: Parts of Flight

1.4. The Ramjet Technical

The F-16 is a single-engine fighter aircraft, used afterburning turbofan engine. These special aircraft engines provide bursts of thrusts that dramatically increase the power output they produce for sprints of extra high speeds. The afterburner is a kind of second combustion chamber that takes the hot gases expelled at the end of the turbofan's cycle and reignites them to produce even more power. The afterburner is located in the **exhaust duct**, behind the turbines. It takes advantage of a key fact: the air leaving the engine still contains about **75% of its original oxygen**.

The major components of an afterburning turbofan and its associated stages are:

- Fan: Captures and compresses incoming air during the intake stage.
- Compressor: Increases air pressure in the compression stage.

- Combustion Chamber: This chamber combines compressed air with fuel for combustion, producing hot, high-pressure gas in the combustion stage.
- Turbine: Extracts energy from the combustion gases to power the compressor and fan at this stage.
- Exhaust ramjet: Expands and directs gases to maximize thrust in the exhaust stage.
- Afterburner: Combusts additional fuel in the post-turbine stage to increase thrust.

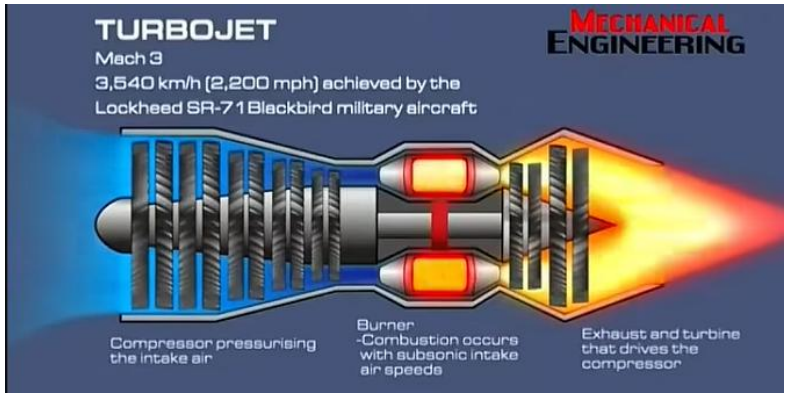

Figure 8: The Process of Ramjet

❖ The afterburning turbofan operate on 4 stages, such as

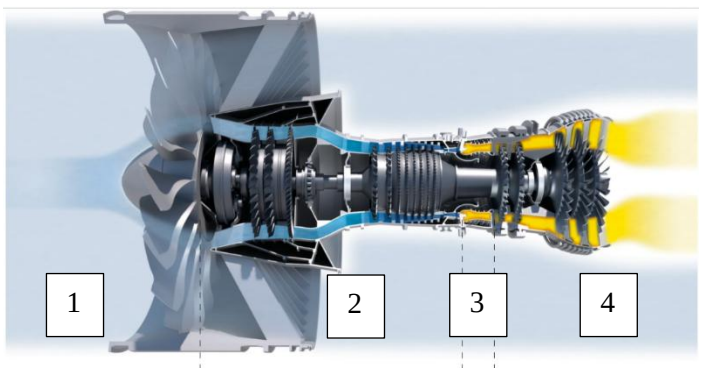

Figure 9: The Turbofan Process

1. **Ingestion**: A turbofan splits air into two paths: a small core flow for combustion and a larger bypass flow (up to 80%) that moves around the center. This bypass air efficiently generates most of the thrust while acting as a "cold sheath" to muffle engine noise.

Temperature variation with altitude:

$$T_1 = T_0 - L \times h$$

T0: Reference temperature at sea level [K]
L: Temperature lapse rate [K/m]
h: Altitude above sea level [m]

Isentropic Flow:

case $P_1 = P_s$

$$P_s = P_0 \left(\frac{T_1}{T_0}\right)^{\frac{\gamma}{\gamma-1}}$$

$$\phi_1 = \frac{P_s}{R_{gas} \times T_1}$$

$$C = \sqrt{(\gamma \times R_{gas} \times T_1)}$$

$$V_1 = \sqrt{\left(2\frac{P_s - P_t}{\phi_1}\right)}$$

$$M_1 = \frac{V_1}{C}$$

When

- Ps: Static Pressure [Pa]
- Pt: Total Pressure [Pa]
- φ: Density function [kg/m3]
- V1: Air speed [m/s2]
- M1: Mach Number
- T1: Temperature inlet [K]
- P1: Pressure inlet [Pa]
- C: Sound speed [m/s2]

2. **Compression**: Inside the engine core, air is squeezed in two stages: first by the low-pressure booster, then by the high-pressure compressor. The high-pressure stage uses "blisk" technology—where blades and disks are made as a single part—to maximize compression efficiency while keeping the engine lightweight.

From 01 to 1 is Isentropic Flow:

$$\frac{T_{01}}{T_1} = \left(1 + \frac{\gamma - 1}{2}\right) \times (M_1)^2$$

and

$$\frac{P_{01}}{P_1} = \left(\frac{\mathrm{T}_{01}}{\mathrm{T}_1}\right)^{\gamma/\,\gamma-1}$$

so

$$\frac{P_2}{P_1} = \frac{1+(\gamma \times M_1^2)}{1+\gamma} \quad \text{and} \quad \frac{P_{02}}{P_{01}} = \frac{1+(\gamma \times M_1^2)}{1+\gamma}$$

$$\frac{T_2}{T_1} = (M_1 \times \frac{P_1}{P_2})\text{^}2 \text{ and } \frac{T_{02}}{T_{01}} = (M_1 \times \frac{P_1}{P_2})\text{^}2$$

Let the following variables be defined:

- T01, P01: Total (stagnation) temperature and pressure at station 1 [K]; [Pa]
- T2, P2: Static temperature and pressure at station 2 [K]; [Pa]
- γ: Ratio of specific heats (Gamma)

3. **Combustion**: In the **combustor**, fuel is sprayed into the compressed air and ignited at roughly **1,700°C**. This extreme heat causes the gas to expand massively, creating a high-energy stream that blasts out of the chamber.

Relation of area and Mach number

$$\frac{A_3}{A_2} = \frac{1}{\mathrm{M}_3}\left[\frac{2}{\gamma+1} \times 1 + (\gamma - 1)\frac{M_3^2}{2}\right]^{\frac{\gamma+1}{2(\gamma-1)}}$$

Compute static temperature and pressure at section 2

Isentropic Flow:

$$\frac{T_3}{T_{02}} = \left(1 + \frac{\gamma - 1}{2}\right) \times (M_3)^2$$

$$\frac{P_3}{P_{02}} = \left(\frac{T_3}{T_{02}}\right)^{\gamma/\gamma-1}$$

$$\frac{T_{03}}{T_3} = \left(1 + \frac{\gamma - 1}{2}\right) \times (M_3)^2$$

$$\frac{P_{03}}{P_3} = \left(\frac{T_{03}}{T_3}\right)^{\gamma/\gamma-1}$$

Speed outlet

$$V_3 = M_3 \times a_3$$

$$that \quad a_3 = sqrt(\gamma \times R_{gas} \times T_3)$$

4. **Expulsion**: The hot gas spins the high- and low-pressure turbines, which use about 60% of the gas's energy to power the engine's compressors and fan. This energy is transferred via two concentric shafts: the outer shaft connects the high-pressure components, while the inner shaft drives the fan and booster. The remaining gas then exits through the ramjet to create residual thrust.

❖ **Engine operating cycle:**

The turbofan engine operates on the Brayton Cycle, a thermodynamic process that describes how a gas (air) is used to create work. It follows four specific steps:

- **Isentropic Compression:** The fan and compressors squeeze the air, increasing its pressure and temperature without adding heat yet.

In this stage, is done on the air to increase its pressure.

➢ Isentropic Relations : $\frac{T_2}{T_1} = \left(\frac{P_2}{P_1}\right)^{\frac{\gamma-1}{\gamma}}$

- **Isobaric Heat Addition:** In the combustor, fuel is added and burned at constant pressure. This is where the energy from the fuel enters the system.

Fuel is burned at constant pressure $P_2 = P_3$significantly increasing the temperature.

➢ Specific Heat Addition during the combustion process

$$q_{in} = \dot{m}C_p(T_3 - T_2)$$

- **Isentropic Expansion:** The hot, high-pressure gas expands through the turbines and the ramjet. This turns heat energy into mechanical work the and kinetic energy.

The high-energy gas expands to produce work. In a turbofan, some of this work powers the compressor, while the rest provides thrust.

➢ Network: $W_{net} = W_t - W_c$

➢ Turbine work: $W_t = \dot{m}C_p(T_3 - T_4)$

- **Isobaric Heat Rejection:** The hot exhaust leaves the engine and cools down in the atmosphere, completing the cycle.

The cycle "closes" as the exhaust air cools back to ambient conditions at constant pressure.

Heat rejection: $q_{out} = \dot{m}C_p(T_4 - T_1)$

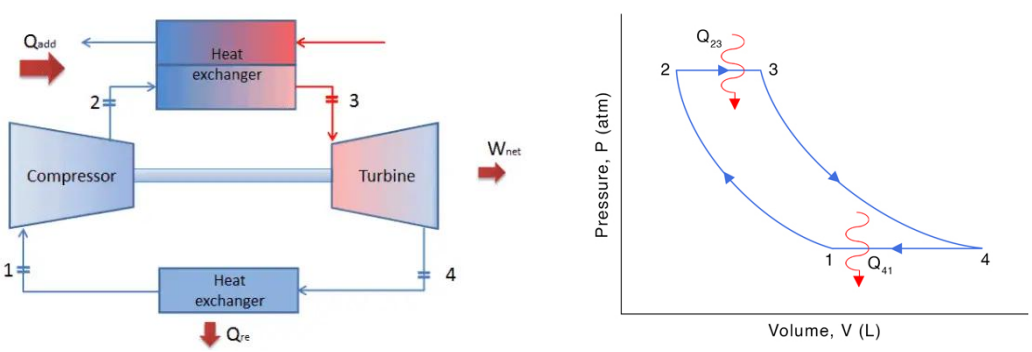

Figure 10: The Engine Cycle

Engine (F100 or F110 afterburning turbofan)

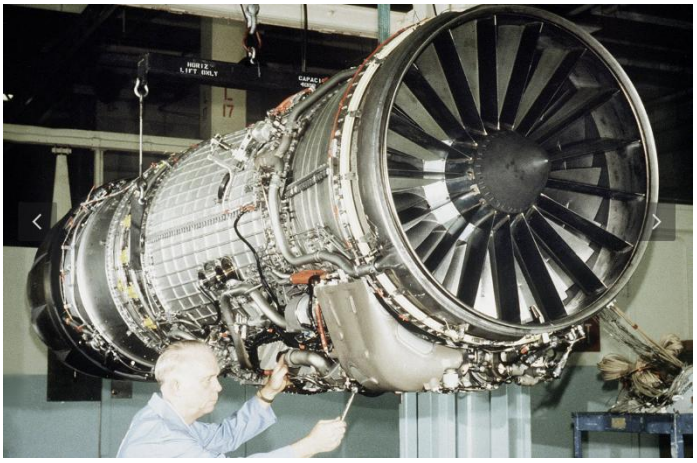

Figure 11:Engine (F100 or F110 afterburning turbofan)

Parameter	Value (approx.)	SI Unit
Type	Low-bypass afterburning turbofan	—
Length	4.7 m	meters
Diameter	1.2 m	meters
Dry weight	1,700 kg	kilograms
Thrust (military power)	106,000 N	Newtons
Thrust (afterburner)	129,000 N	Newtons
Bypass ratio	0.36	dimensionless
Fuel flow (afterburner)	~2.5 kg/s	kilograms per second
Specific fuel consumption (military)	~0.8 kg/(N·h)	kilograms per Newton-hour

Table 1: Some detail of machine F100 of F16

The F-16's jet machine is a compact, high-power turbofan about 5 m long and 1.7 tones heavy, producing up to 129 kN of thrust with afterburner. This thrust gives the aircraft supersonic speed beyond Mach 2, extreme agility, and the ability to climb vertically — all while burning fuel at several kilograms per second in full afterburner

2. Material

When an aircraft breaks the sound barrier, air friction creates a "thermal thicket"—a wall of heat that can melt standard materials. Titanium is the primary material used to overcome this because it maintains its structural integrity at temperatures that would cause aluminum to turn into a "warm noodle."

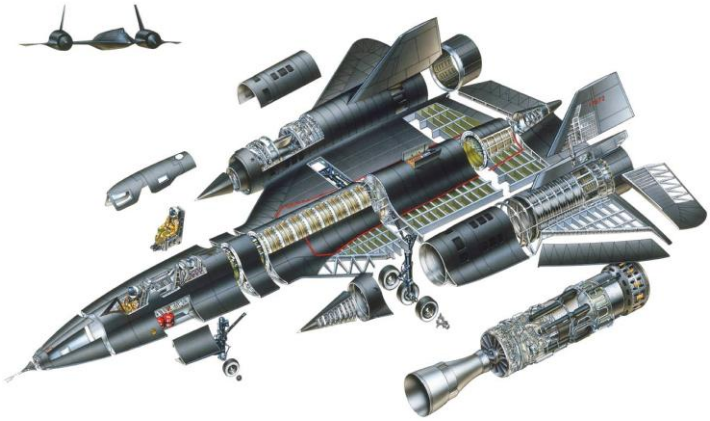

Figure 12: Type of Material

This Titanium is the "Supersonic Metal". Titanium is essentially the middle ground between aluminum (light but weak in heat) and steel (strong but too heavy).

- **Thermal Stability:** Aluminum loses its strength around 135°C, whereas titanium alloys can operate reliably up to 550°C to 600°C.
- **Expansion Control:** Titanium has a very low Coefficient of Thermal Expansion. It doesn't "grow" or warp as much as other metals when it gets hot, which is vital for keeping the aircraft's shape aerodynamic at Mach 3.
- **Strength-to-Weight:** It is as strong as steel but 45% lighter, allowing supersonic jets to carry enough fuel to maintain high speeds.

The SR-71 Blackbird: A Titanium Case Study

The SR-71 is the most famous example of titanium engineering. Because it flew at Mach 3.2, its skin reached temperatures over 315°C (600°F).

- **The "Leaking" Design:** The titanium panels were designed to fit loosely on the ground. Engineers left gaps between them because they knew the metal would expand by several inches during flight. This meant the plane actually leaked fuel on the runway until it got hot enough for the titanium to "grow" and seal the tanks.
- **Titanium Tools:** Standard cadmium-plated tools would react with the titanium and cause it to become brittle (a process called embrittlement). Workers had to use specialized titanium wrenches to build the plane.

- **Sourcing:** At the time (the 1960s), the US didn't have enough titanium. They actually set up "shell companies" to buy the raw ore from the Soviet Union—the very country they were building the plane to spy on.

Here is the common titanium alloys used in aircraft, with a clear explanation of engineers choose each one for specific airplane parts to improve strength, heat resistance, fatigue life, and corrosion resistance.

2.1. Type of Titanium

Type of Titanium	Reasons	Application
Ti-6Al-4V (Grade 5)	• Very high strength-to-weight ratio • Excellent fatigue resistance • Good high-temperature performance (up to ~400°C)	• Jet engine compressor blades • Engine casings • Wing and fuselage structural parts • Landing gear components
Ti-6Al-2Sn-4Zr-2Mo (Ti-6242)	• Maintains strength at higher temperatures than Ti-6Al-4V • Excellent creep resistance	• Jet engine compressor discs • Engine hot-section structures • Supersonic aircraft components
Ti-5Al-2.5Sn (Grade 6)	• Performs well at very low temperatures • Good weldability • Moderate strength	• Fuel tanks • Airframe parts exposed to cold environments • Aerospace tubing
Beta Titanium Alloys (e.g., Ti-10V-2Fe-3Al)	• Extremely high strength • Can be heat-treated for different properties • Good formability before heat treatment	• Landing gear • High-load fasteners • Structural fittings
Commercially Pure Titanium (Grades 1–4)	• Excellent corrosion resistance	• Hydraulic lines • Exhaust ducts

	• Very ductile and easy to form • Lower strength than alloys	• Chemical and corrosion-prone areas

Table 2: The Type of Titanium

2.2. Challenging

While titanium is amazing, it is 10x more expensive than aluminum and incredibly difficult to cut and weld. This is why modern supersonic companies like Boom Supersonic are moving toward Carbon Fiber Composites, which are easier to manufacture into complex shapes but still offer high heat resistance when paired with specialized resins.

Building a supersonic aircraft with Titanium requires a different set of engineering rules compared to standard aluminum planes. Because Titanium is exceptionally hard, heat-resistant, and expensive, it must be "used" through specific manufacturing and design philosophies.

3. Evaluation and Analysis

For Analysis and Evaluation, we will use some software which are useful and most popular in the aerodynamic. They have an efficiency and good commands for user set conditionals, find the variable, plot graphs, and etc. We will set the family of the application for you; it will be helped you easier and more convivence to understand with this path.

We derivate it to 3 groups like **CAE (Computer-Aided Engineering)**, **CAD** (**Computer-Aided Design), and CFD (Computational Fluid Dynamics**).

3.1. CAE (Computer-Aided Engineering)

For **CAE**, we have 5 software such as Microsoft Excel, MATLAB, GNU Octave, Python code and C++ code. Those app would allow you to calculation, simulation, numerical computing, modeling, visualization and etc.

Figure 13: The CAE applications

For this path, we will choose the MATHLAB to run with these problems.

Moreover, this part we have 3 point that we have to consider and analyses like:

- **The link area ratio and Mach number**

Determine the relation of area and Mach number, it is the first step that we should find it. After, we get the A'/A ratio, we are able to design the sample of ramjet.

- **The result**

Putting the data, evaluate to find the theory result. Some point, we use the data from the internet like pressure of tube and temperature at 5000m compare with sea-level, since we don't have enough equipment to measure it.

3.1.1.The Link Area Ration and Mach Number

```
%%
Aratio= 3;
g=1.4;

fcn = @(M) (1./M).*( (2/(g+1)) .* (1+(g-1)/2*M.^2) ) .^ ((g+1)/(2*(g-1))) -
Aratio;

Mv = linspace(0, 4);
idx = find(diff(sign(fcn(Mv))))
```

idx = *1×2*

5 66

```
for k = 1:numel(idx)
    [Mc(k),fv(k)] = fzero(fcn, Mv(idx(k)));
end
figure
plot(Mv, fcn(Mv), '-b', Mc, zeros(size(Mc)),'sr')
grid
text(Mc,zeros(size(Mc)),compose(' \\leftarrow %.4f',Mc))
% Mc = Mach numbers that make the area ratio equal to 4 (both subsonic and
supersonic solutions).
% fv = residuals of the function at those Mach numbers (should be ≈ 0).
```

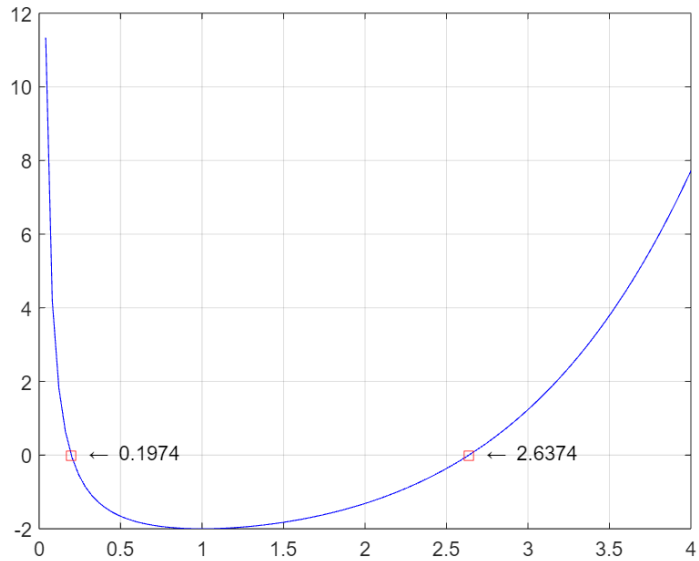

Table 3:Mach number M1 & M3

3.1.2.The Result

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I. Stagnation Inlet Flow

II. Ramjet 1 : Subsonic Flow ---> Sonic Flow

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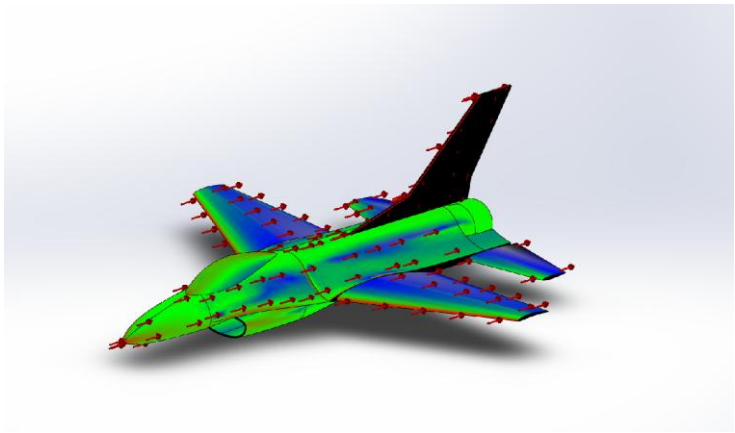
Super Sonic Flight



I. Stagnation Inlet Flow

```
% Pt is Total pressure
Pt = 52529;     % Pt = 52529 Pa
% Hight of Sea-level
h = 5000;

% Constants
Gamma = 1.4;
Rgas = 287;     % J/(kg*K)
g = 9.81;       % m/s^2
L = 0.0065;     % K/m
T0 = 288.15;    % Sea-level temperature [K]
P0 = 101325;    % Sea-level pressure [Pa]

% Standard atmosphere (troposphere)
T1 = T0 - L*h;
% Ps is Static pressure
Ps = P0 * (T1/T0)^(g/(L*Rgas));
phi = Ps / (Rgas*T1);
C = sqrt(Gamma * Rgas * T1);
P1 = Ps;
% Air Speed
V1 =sqrt(2*((-Pt+Ps))/phi);
% Velocity
M1 = V1/C;
% Display
fprintf('Inlet: M1=%.4f, V1=%.4f m/s, P1=%.4f Pa, T1=%.4f K' , ...
    M1, V1, P1, T1);
```

```
Inlet: M1=0.1974, V1=63.2672 m/s, P1=54002.0229 Pa, T1=255.6500 K
```

II. Ramjet 1: Subsonic Flow ---> Sonic Flow

```
% Stagnation Inlet
T01 = T1 *(1+((Gamma-1)/2)*M1^2);
P01 = P1 * (T01/T1)^(Gamma/(Gamma-1));
P2 = P1*((1+Gamma*M1^2)/(1+Gamma));
T2 = T1 / (1 + ((Gamma-1)/2));
% Sonic Flow
M2 = 1;

% Speed at in the Ramjet 1
V2 = M2*C ;
```

```
% No Heat at M2=1
P02 = P01 * ((1 + Gamma * M1^2) / (1 + Gamma));
T02 = T2 * (1 + (Gamma-1)/2 * M2^2);

% Static Pressure
Delta_P_1 = P1 - P2;

% Diplay
fprintf('Throat: M2=%.4f, V2=%.4f m/s, P2=%.4f Pa, T2=%.4f K, T02=%.4f K,
, P02=%.4f Pa, ΔP1=%.4f Pa\n', ...
    M2, V2, P2, T2, T02, P02, Delta_P_1);
```

```
Throat: M2=1.0000, V2=320.4999 m/s, P2=23728.3620 Pa, T2=1266.6593 K,
T02=255.65 K, P02=24381.9348 Pa, ΔP1=30273.6609 Pa
```

III. Ramjet 2: Sonic Flow ---> Super Sonic Flow

```
% Area Ratio
A2_A3 = 3;

% Solve for M3 using area-Mach relation (numerical root)
area_mach = @(M) (1/M)*((2/(Gamma+1)*(1 + ((Gamma-
1)/2)*M^2))^((Gamma+1)/(2*(Gamma-1)))) - A2_A3;
M3_guess = 2.6374;
M3 = fzero(area_mach, M3_guess);

% Compute static temperature and pressure at section 3
T3 = T02 * (1 + ((Gamma-1)/2)*M3^2);
P3 = P02 * (T3/T02)^(-Gamma/(Gamma-1));

T03 = T3 * (1 + (Gamma-1)/2 * M3^2);
P03 = P3 * (1 + (Gamma-1)/2 * M3^2)^(Gamma/(Gamma-1));

% Speed of sound and velocity
a3 = sqrt(Gamma * Rgas * T3);
V3 = M3 * a3;

% Diplay
fprintf('Section 3: M3=%.4f, V3=%.4f m/s, P3=%.4f Pa, T3=%.4f K, T03=%.4f K,
P03=%.4f Pa\n', ...
    M3, V3, P3, T3, T03, P03);
```

```
Section 3: M3=2.6374, V3=1307.1148 m/s, P3=609.2323 Pa,T3=611.3084 K,
T03=1461.7558 K, P03=12880.5321 Pa
```

IV. Display Results

```
%% Store results in cell array
ResultsCell = {
    'Inlet',     M1, V1, P1/1000, T1;
    'Throat',    M2, V2, P2/1000, T2;
    'Outlet',    M3, V3, P3/1000, T3;
};
% Convert to table
ResultsTable = cell2table(ResultsCell, ...
    'VariableNames', {'Section', 'MachNumber', 'Velocity_mps', 'Pressure_KPa',
'Temperature_K'});
% Display
disp(ResultsTable)
```

Section	Mach Number	Velocity (m/s)	Pressure (Kpa)	Temperature (K)
{'Inlet'}	0.1974	63.267	54.002	255.65
{'Throat'}	1	320.5	23.728	213.04
{'Outlet'}	2.6374	1221.5	0.60923	611.31

Table 4: The result values

```
% Define section labels and positions
sections = {'Inlet', 'Throat', 'Outlet'};
x = 1:3; % Position index for plotting

% Define flow properties
Mach = [0.1974, 1, 2.6374];
Velocity = [63.267, 320.5, 1221.5];
Pressure = [54.002, 23.728, 1.1532]; % in kPa
Temperature = [255.65, 1266.7, 533.85]; % in K

% Create figure with subplots
figure;

subplot(2,2,1)
plot(x, Mach, '-o', 'LineWidth', 2)
xticks(x); xticklabels(sections)
ylabel('Mach Number')
title('Mach Number vs Section')
grid on

subplot(2,2,2)
```

```
plot(x, Velocity, '-o', 'LineWidth', 2)
xticks(x); xticklabels(sections)
ylabel('Velocity (m/s)')
title('Velocity vs Section')
grid on

subplot(2,2,3)
plot(x, Pressure, '-o', 'LineWidth', 2)
xticks(x); xticklabels(sections)
ylabel('Pressure (kPa)')
title('Pressure vs Section')
grid on

subplot(2,2,4)
plot(x, Temperature, '-o', 'LineWidth', 2)
xticks(x); xticklabels(sections)
ylabel('Temperature (K)')
title('Temperature vs Section')
grid on
```

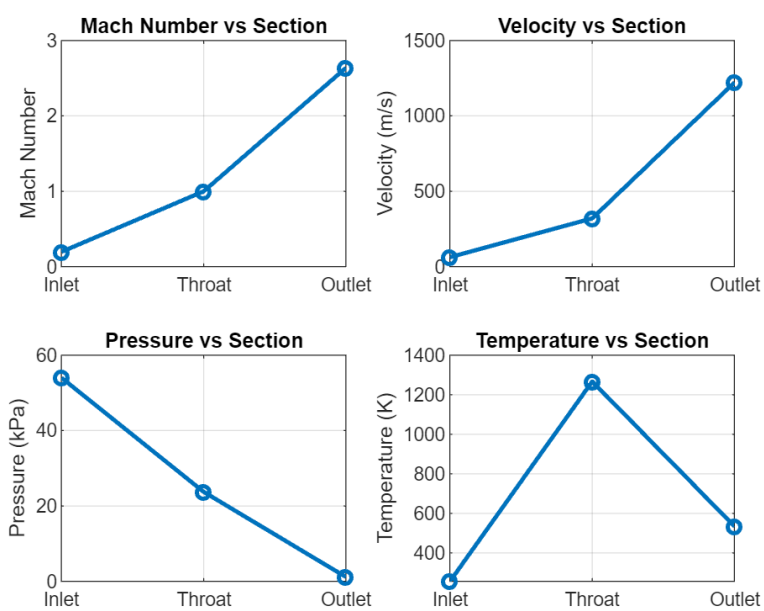


Table 5:The value at each section

3.2. CAD (Computer-Aided Design)

CAD is the software that let engineering draw anything that they want to create or build it to real life object. If accept user to draw 2D, 3D, simulation, and **CAM** (Computer-Aided Manufacturing) too. Mostly you will see the engineering would work and cover on this software especially Mechanical Engineering, Aerospace Engineering, and another field.

In this part, I give you few software that can run with CAD and more popular like SOLIDWORK, AutoCAD, Fusion 360, Blender, and more.

For this path, we use the SOLIDWORKS to draw the concept of mechanic.

Figure 14: The CAD applications

- **Airplane**

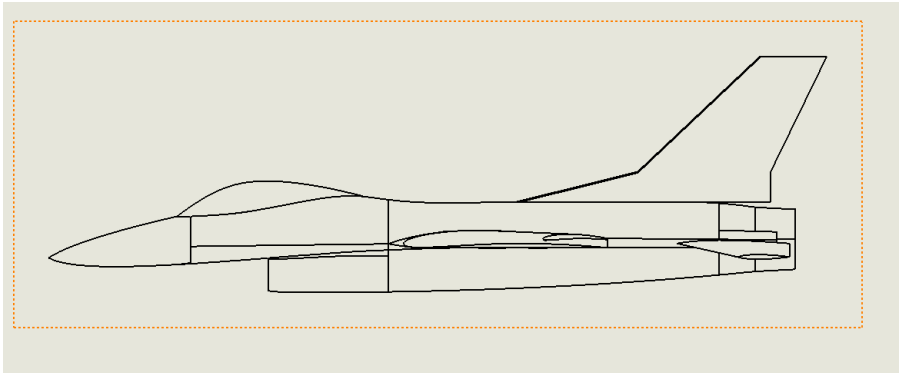


Figure 15: Airplane

The geometry that we focus are:

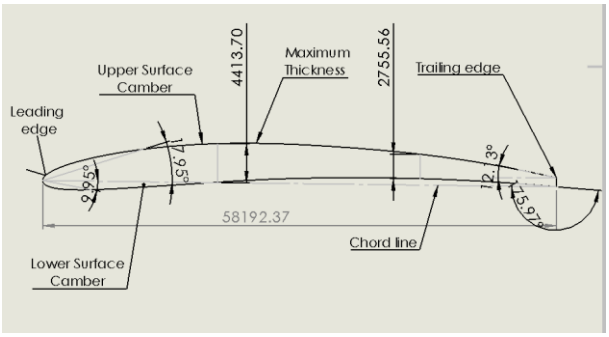
- **Main wing:** Features cambered upper and lower surfaces, optimized for lift generation during takeoff and cruise.
- **Tail airfoil:** Designed with precise angles and thickness placement to ensure pitch stability and control authority.
- **Ramjet integration:** Positioned at the rear, shaped to accelerate exhaust flow and support thrust generation.

The reason that we choose this geometry:

- **Aerodynamic efficiency**: The fuselage and wing contours were shaped based on velocity and pressure field simulations, ensuring smooth flow and minimal separation.
- **Lift and stability**: Wing camber and airfoil geometry were tuned to produce high Cl during takeoff and maintain low Cd during cruise.
- **Thermal and structural considerations**: Ramjet and leading-edge geometries reflect cooling effects and pressure gradients observed in CFD results.
- **Control surfaces**: Tail airfoil angles were selected to delay boundary layer transition and avoid

trailing edge separation, ensuring responsive pitch control.

- **Airfoil**



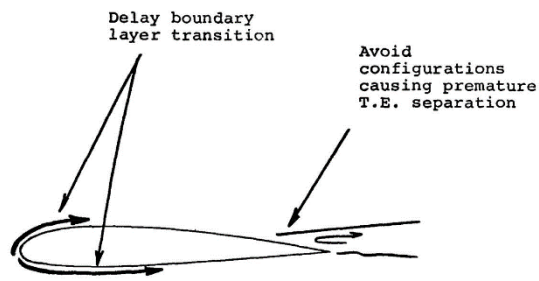


Figure 16: Airfoil

Why we choose the angle of leading edge is small, because angle is affected the Cl and Cd.

Geometric Parameters:

The airfoil design includes precise dimensional and angular features that influence aerodynamic performance:

- **Chord length**: 58,192.37 mm the baseline for calculating aerodynamic coefficients.
- **Maximum thickness**: 4,413.70 mm, located 2,755.56 mm from the leading edge — this affects pressure distribution and boundary layer behavior.
- **Leading edge angle**: 9.25° a small angle chosen to reduce flow separation and delay boundary layer transition.
- **Lower surface angle**: 19.95° contributes to pressure recovery and drag control.
- **Trailing edge angles**: 12.3° (upper) and 15.97° (lower) — designed to avoid premature separation and ensure smooth flow exit.

These geometric choices are critical for balancing lift and drag, especially in supersonic or transitional flow regimes.

Flow Behavior and Design Rationale:

The airflow diagram shows how geometry influences aerodynamic stability:

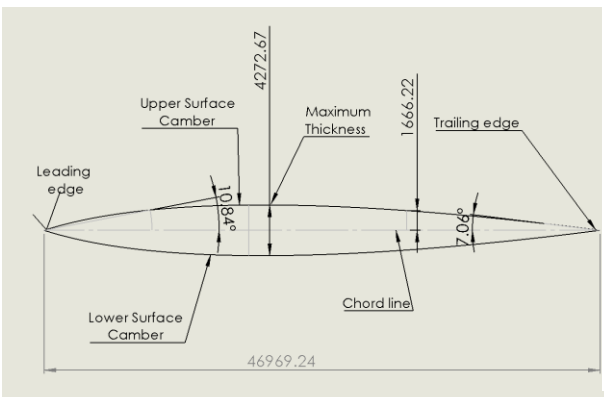
- **Delayed boundary layer transition**: The gentle leading edge and camber design help maintain laminar flow longer, reducing skin friction drag.

- **Avoiding trailing edge separation**: Proper trailing edge angles prevent flow detachment, which would otherwise increase pressure drag and reduce lift.

Impact on Aerodynamic Coefficients

- **Lift coefficient (Cl)**: A small leading-edge angle and smooth camber promote favorable pressure gradients, increasing Cl during takeoff or positive angle of attack.
- **Drag coefficient (Cd)**: Streamlined geometry and delayed separation reduce Cd, improving efficiency during cruise.

- ## Tail airfoil



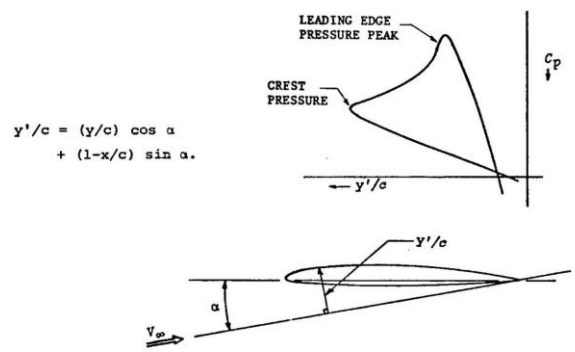


Figure 17:Tail airfoil

- **Leading-edge angle (10.84°) → Cp peak and control authority:** A sharper nose intensifies local deceleration, raising Cp near the front. For a tail surface, this strengthens pitch response at small α, improving trim and stability sensitivity.
- **Thickness placement (early) → pressure crest and lift modulation:** Positioning maximum thickness close to the leading edge accelerates the upper-surface flow sooner, lowering pressure and producing an early crest. This shapes the lift curve slope and helps the tail generate corrective moments efficiently without large α.
- **Trailing-edge angle (7.06°) → delayed separation and low drag:** A gentle TE angle eases pressure recovery, reducing adverse gradients, keeping the boundary layer attached, and minimizing induced pressure drag—important for predictable control forces.
- **Angle-of-attack mapping → pressure feature migration:** As α increases, the transform shifts effective chordwise coordinates, moving the Cp peak and crest downstream on the projected chord. This explains changes in lift and pitching moment with α and why tail effectiveness scales smoothly with small deflections.
- **Tail role (stability over lift):** Geometry emphasizes linear, low-hysteresis Cp behavior rather than maximum Cl. The result is consistent aerodynamic damping and precise pitch control, supporting longitudinal stability across operating conditions.

- **Ramjet**

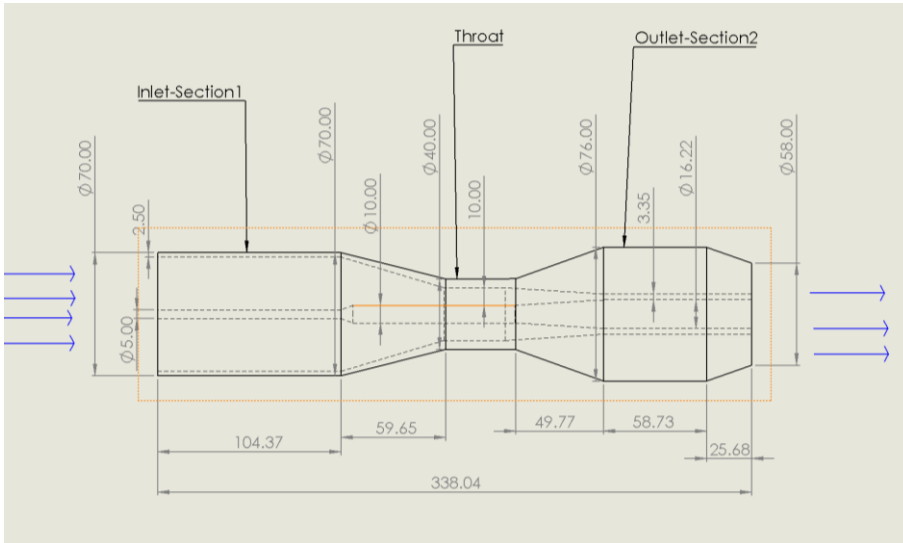


Figure 18:Ramjet

The ramjet system consists of two main sections with a converging-diverging profile:

- **Inlet Section 1**

Length: 104.37 mm

Diameter: 70.00 mm

Hole diameter: 60.00 mm

Mach number transition: M1=0.1974 (subsonic)

- **Throat**

Diameter: 40.00 mm

Length: 10.00 mm

Acts as the critical flow point where velocity reaches sonic conditions (M=1)

- **Outlet Section 2**

Length: 58.73 mm

Diameter: 76.00 mm

Mach number transition: M2=1 to supersonic

Area ratio: A1/A2=3, A2/A3=3

Turbine Shaft Functionality:

At the center of the Ramjet lies a **rotating shaft** or **blade turbine**, which serves a dual purpose:

- **Energy conversion**: Transforms air force into mechanical energy, potentially driving auxiliary systems or enhancing thrust.

- **Velocity enhancement**: By extracting energy from the flow, the turbine can modulate downstream velocity and pressure, improving propulsion efficiency.

Flow Behavior and Simulation Insights:

- **Pressure gradient**: High static pressure at the inlet (red zones) transitions to low pressure at the outlet (blue zones), confirming effective expansion.
- **Velocity increase**: Streamlines and vector arrows show smooth acceleration through the throat, with no signs of separation or turbulence.
- **Thermal and pressure stabilization**: Graphs indicate rapid changes in temperature, pressure, and velocity during startup, followed by steady-state behavior.

3.3. CFD (Computer Fluid Dynamics)

CFD focused on simulating fluid flow, heat transfer, and related physical phenomena using numerical methods. **CFD** tools solve the governing equations of fluid mechanics (Navier–Stokes equations) to predict flow behavior around objects like aircraft, cars, turbines, or pipes.

CFD have many software that useful and have better tool for engineering to analysis, set condition, and find the result like ANSYS, Open FOAM, MATHLAB, and etc.

Figure 19:The CFD applications

For this path, we will use Ansys Fluent to analyze the sample to get compare with the theories.

3.3.1. The Airplane

- **Velocity**

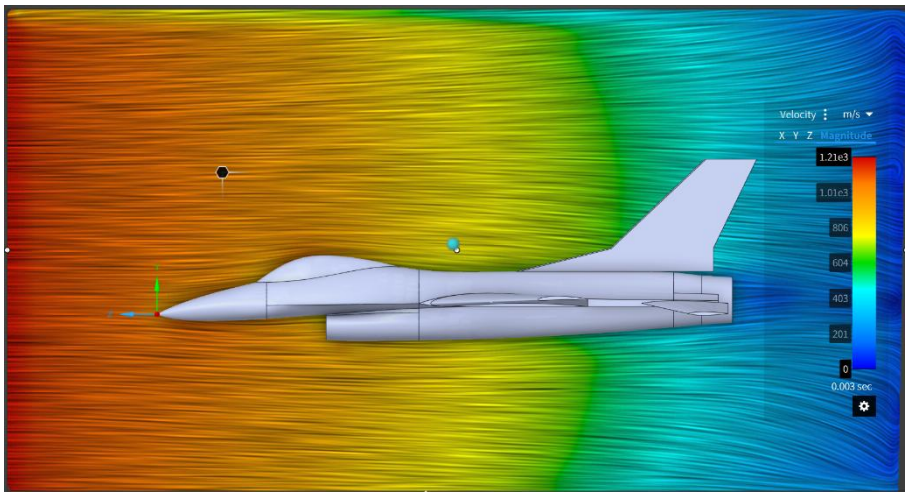


Figure 20: The speed of airplane-simulation

- This figure shows the airplane's velocity field. The color bar ranges from blue (low) to red (high). Each color has range value 0, 201, 403, 604, 806, 1010, 1210 map to clearly to blue/ light blue/ sky/ green/ yellow/ orange/ red. Streamlines indicate airflow direction.
- **The nose** splits the incoming flow into upper and lower paths and sits in a high-velocity zone. The nose of plane is red, it means that face will create the **higher heat** by air friction, **Shock flow**. Moreover, it is a supersonic case, so be carefully at the nose of plane.
- **The wings** lie in medium-velocity regions (yellow/green) where velocity gradients and attached streamlines support lift Cl and influence drag Cd. Lower velocities appear in the wake downstream, consistent with momentum deficit and higher pressure drag.

Region	**Velocity Range (m/s)**	**Color**	**Function**	**Risk**
Nose	1010–1210	Red	Flow splitting, stagnation heating	Thermal stress, shock
Upper wing	604–806	Yellow/Green	Lift generation via pressure drop	Separation risk
Lower wing	403–604	Sky/Green	Pressure recovery, drag balance	Shock formation
Wake	0–201	Blue	Momentum deficit, pressure drag	Turbulence, instability

Table 6:The affection of higher speed

Why speed can affect the Cd and Cl because:

$$U(y) = U_{inf} \times [0.75 - 0.25 * cos\left(pi * \frac{y}{H}\right)]$$

$$Cd = \frac{D}{\left(0.5 * rho * U_{inf}^2 * C\right)}$$

$$Cl = \frac{L}{\left(0.5 * rho * U_{inf}^2 * C\right)}$$

- d is Integrate momentum deficit
- L is Integrate momentum lifting
- U_{inf} is speed inlet of air
- Rho is the air density

By using Momentum Equation:

$$\sum F = \frac{d}{dt}\int Rho * VdV + \int Rho * V * (V.n)dA$$

- ∑F: Sum of external forces = FB + FS
- CV: Control volume (region of fluid we're analyzing)
- CS: Control surface (boundary of the control volume)
- V: Velocity vector
- n: Outward normal vector on the surface

Steady flow, we will get:

$$D = \rho \cdot \int_0^H (U\infty - U(y)) \cdot U(y)\, dy$$

$$L = \rho \cdot \int_0^H V(y) \cdot U(y)\, dy$$

In this case, it doesn't steady flow, so the equation would be more complex and harder to solve. It is just a basic concept to find Cd and Cl for student to learn and know how it works in Gas mechanic.

- Pressure

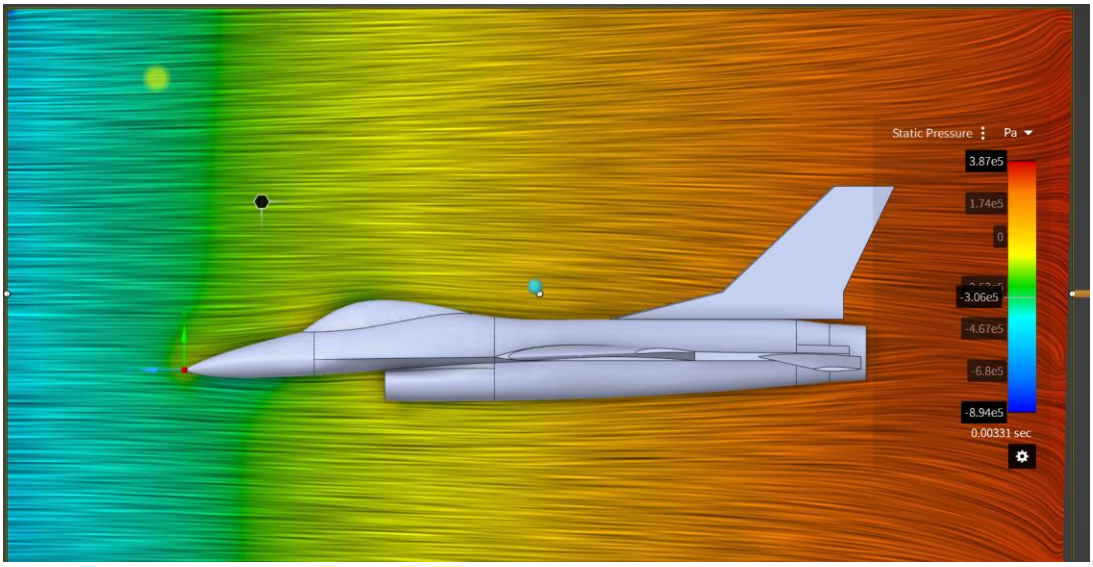


Figure 21:The Static pressure of airplane-simulation

The static pressure field around the aircraft is depicted using a color gradient from −8.496 kPa (blue, low pressure) to +3.496 kPa (red, high pressure):

- **Low-pressure regions (blue):** These zones are concentrated near the nose and lower fuselage. The reduced pressure contributes significantly to pressure drag and affects the overall aerodynamic force balance. These areas are critical for understanding flow acceleration and potential separation.
- **Medium-pressure regions (yellow):** Located over the upper wing surfaces and aft fuselage; these zones play a key role in lift generation. The pressure gradient between the upper and lower wing surfaces supports upward force, while the aft fuselage pressure helps shape the wake and influences stability.
- **High-pressure regions (orange):** Found around the fin, tail, and tailplane; these areas experience elevated pressure (up to 1740 Pa) due to their elevated position relative to the main body. This pressure increase is consistent with flow compression and stagnation effects at the rear control surfaces.
- **Flow Behavior:** The streamline pattern indicates smooth and continuous airflow around the aircraft. The flow appears well-attached, with no major signs of separation or turbulence in the visualized snapshot. This suggests that the aircraft geometry is well-optimized for the current flight condition.

- **Velocity, Static Pressure, and Temperature**

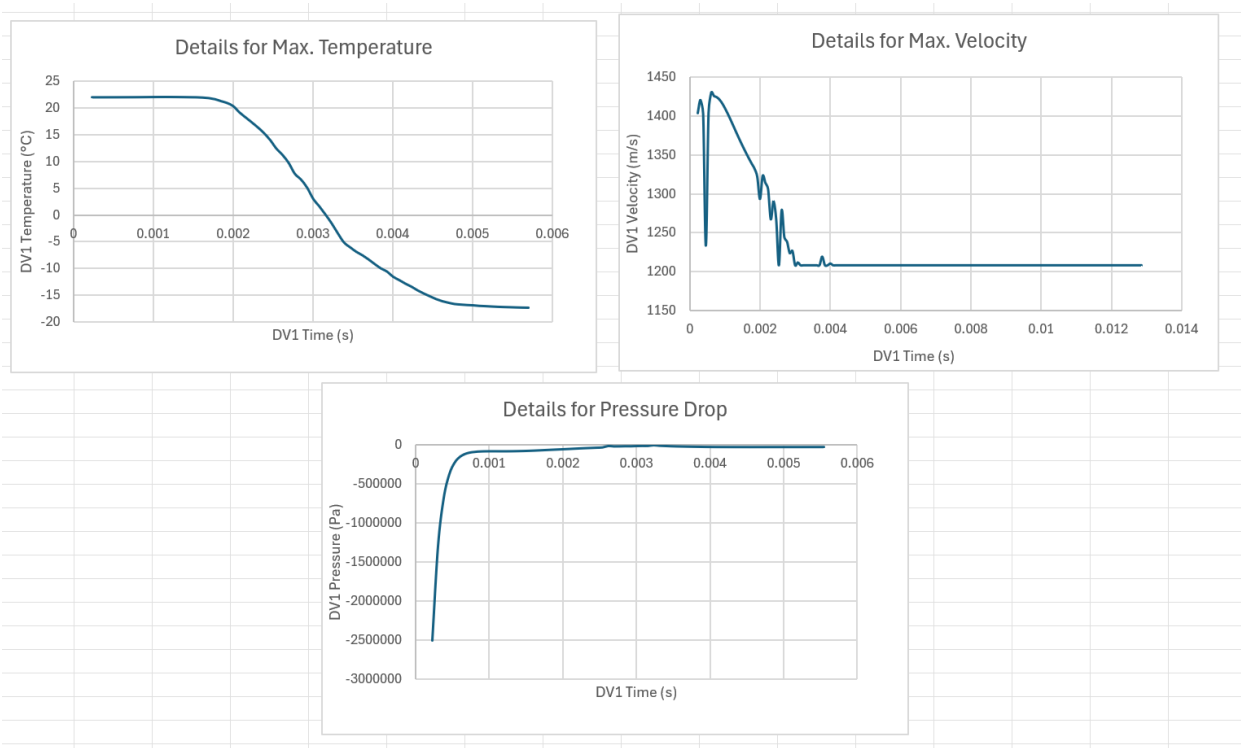


Table 7: The velocity, Static pressure, and Temperature of plane

Velocity Analysis:

- **Initial peak velocity:** Approximately 1400 m/s, indicating rapid acceleration of the airflow as it encounters the aircraft. It happens when the plane takes off the land with specific angle maybe range from 5 to 15 degree. It will make bigger Cl and lower Cd that lift the plane up to the sky.
- **Stabilized velocity:** Approximately 1200 m/s, reflecting the establishment of steady supersonic flow conditions after initial transients. The plane is flied nearly 0 degree and it allows plane run with the stable speed.

This behavior aligns with expected supersonic inlet conditions and the presence of shock waves that adjust the flow velocity.

Temperature Variation:

- **Initial temperature:** Approximately 25°C, representing ambient or baseline conditions.
- **Final temperature:** Approximately −20°C, indicating significant cooling due to convective heat transfer and expansion effects in high-speed flow regions. It colder because the sea-level when the plane flight, it is an importance case that can reduce the heat of machine, Ramjet and plane

This temperature drop is important for thermal management considerations, as it affects material properties and structural integrity.

Pressure Drop Dynamics:

- **Maximum pressure drops:** Approximately −2,500,000 Pa, a sharp decrease associated with shock wave formation and rapid expansion in the supersonic flow.
- **Stabilization:** Pressure levels off after the initial transient phase, indicating flow equilibrium.

Managing these pressure fluctuations is vital to prevent structural damage and maintain aerodynamic stability and the static pressure is very small if comparing it with the input pressure.

Conclusion

- **Lift and drag forces:** Derived from pressure gradients and momentum conservation, these forces dictate aircraft performance and fuel efficiency.
  - High nose velocity → need for heat-resistant materials
  - Wake velocity → fairing and tail shaping to reduce Cd
- **Thermal management:** Critical in regions experiencing rapid temperature changes to ensure material durability and prevent thermal fatigue.
- **Design optimization:** Pressure and velocity maps guide aerodynamic shaping and material selection to enhance performance and safety.

For resistance with lower temperature, higher pressure, and higher velocity, engineering could be considered, analyzed, and chosen the right material that fit the environment.

3.3.2. The Air Foil

- **Velocity**

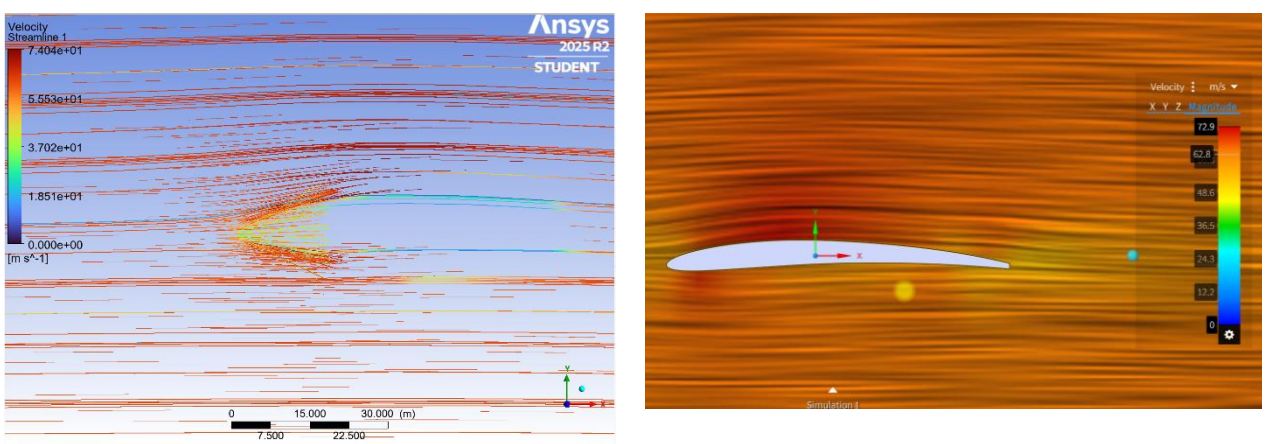


Figure 22:The speed of airfoil-simulation

- **Acceleration over the upper surface**: The flow speeds up significantly, as shown by red and yellow regions in the velocity contour. This increase in velocity contributes to a pressure drop, which supports lift generation.
- **Stagnation at the leading edge**: Velocity drops sharply at the front of the airfoil, marking the stagnation point where the incoming flow momentarily halts before splitting into upper and lower paths.
- **Attached downstream flow**: Beyond the leading edge, the streamlines remain smooth and attached to the surface, indicating stable aerodynamic behavior with minimal flow separation under current operating conditions.

- ## Static Pressure

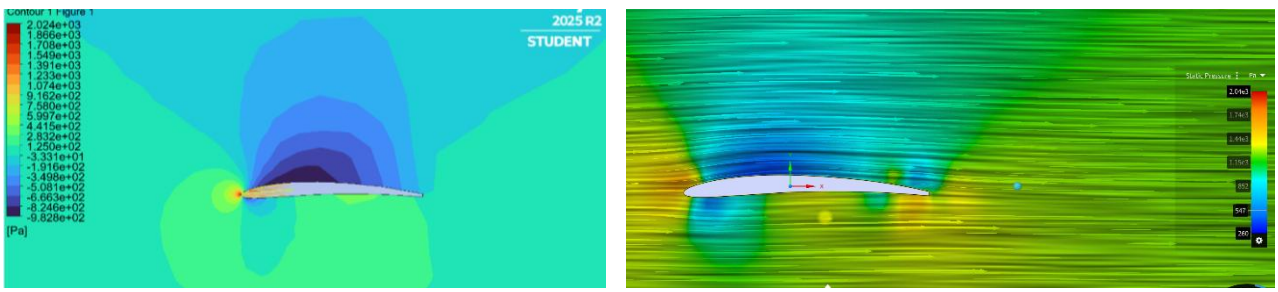


Figure 23: The static pressure of airfoil-simulation

- **High pressure at the leading edge**: The stagnation point causes a buildup of pressure, reaching approximately 2024 Pa due to flow deceleration.
- **Pressure drops along the upper surface**: As the flow accelerates, pressure decreases over the upper surface, forming a low-pressure region that enhances lift.
- **Balanced pressure on the lower surface**: The pressure remains relatively higher, contributing to the net upward force and aerodynamic stability.

- ## Cd (Coefficient Drag) and Cl (Coefficient Lifting)

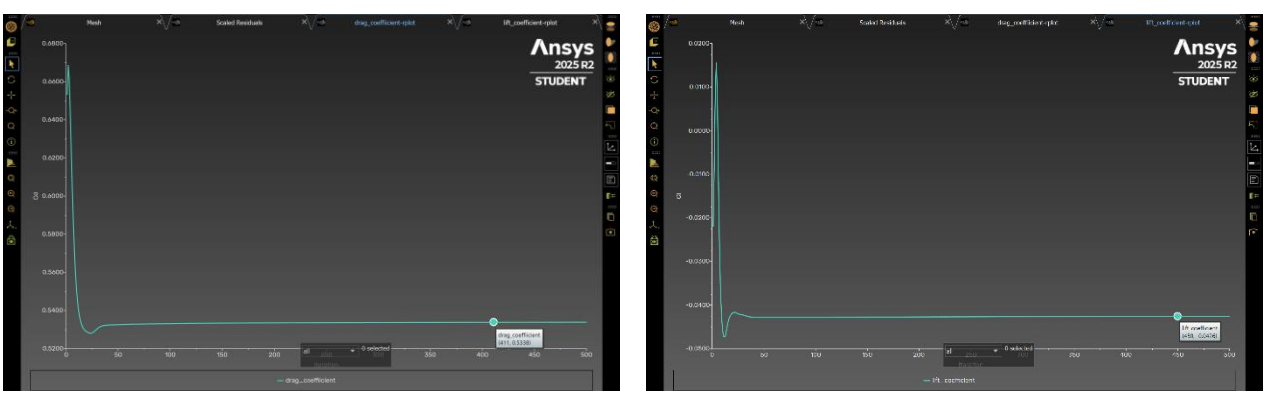


Table 8:The Cd and Cl of airfoil

Drag Coefficient (Cd)

- The final drag coefficient obtained from the simulation is **Cd = 0.5338**, which stabilized after initial fluctuations.
- This value reflects the **resistance force** acting opposite to the flow direction due to pressure and viscous effects on the body surface.
- The convergence behavior suggests that the simulation reached a **steady-state solution**, indicating reliable results.
- A Cd of 0.53 is typical for bluff bodies or fuselage-like geometries with moderate streamlining. If your geometry includes sharp transitions or flow separation zones, this value is physically consistent.

Lift Coefficient (Cl)

- The final lift coefficient is **Cl = 0.0426**, which is relatively low.
- This implies that the body generates **minimal vertical force**, which may be expected if the geometry lacks camber or angle of attack.
- The low Cl also suggests that the flow remains mostly symmetric, with limited circulation or pressure differential between upper and lower surfaces.

Note: This case is happened when the air plane is been flighting in the sky in the each high of sea level.

3.3.3. The Tail Air Foil

- **Velocity**

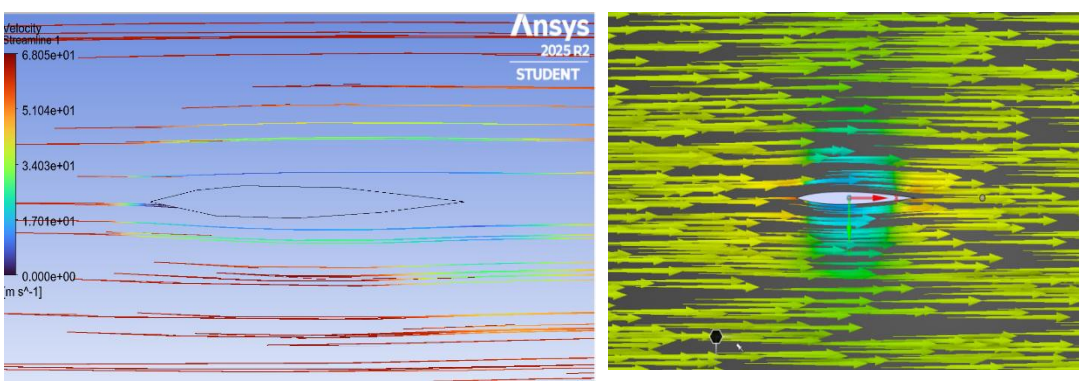


Figure 24:The speed of tail airfoil-simulation

- **Streamline behavior:** The streamlines show smooth flow paths around the airfoil, confirming stable aerodynamic conditions.
- **Velocity gradient:** The color gradient from blue to yellow represents increasing velocity, with yellow indicating faster flow over the upper surface.
- **Lift-related flow:** The upper surface experiences higher velocity, which corresponds to lower static pressure — a classic condition for lift generation.

- **Lower surface flow:** Slower velocity and higher pressure are observed on the lower surface. However, in this case, the lift coefficient remains low, suggesting a minimal net pressure difference across the airfoil.

- **Static Pressure**

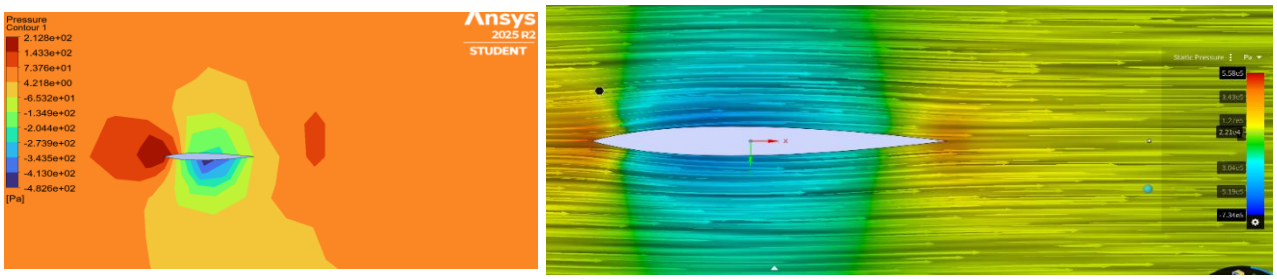


Figure 25:The pressure of tail airfoil-simulation

- **High-pressure zones**: Red regions are concentrated near the leading edge and lower surface, indicating stagnation where airflow decelerates and pressure builds.
- **Low-pressure zones**: Blue regions appear on the upper surface, where the flow accelerates and pressure drops, consistent with Bernoulli's principle.

- **Cd (Coefficient Drag) and Cl (Coefficient Lifting)**

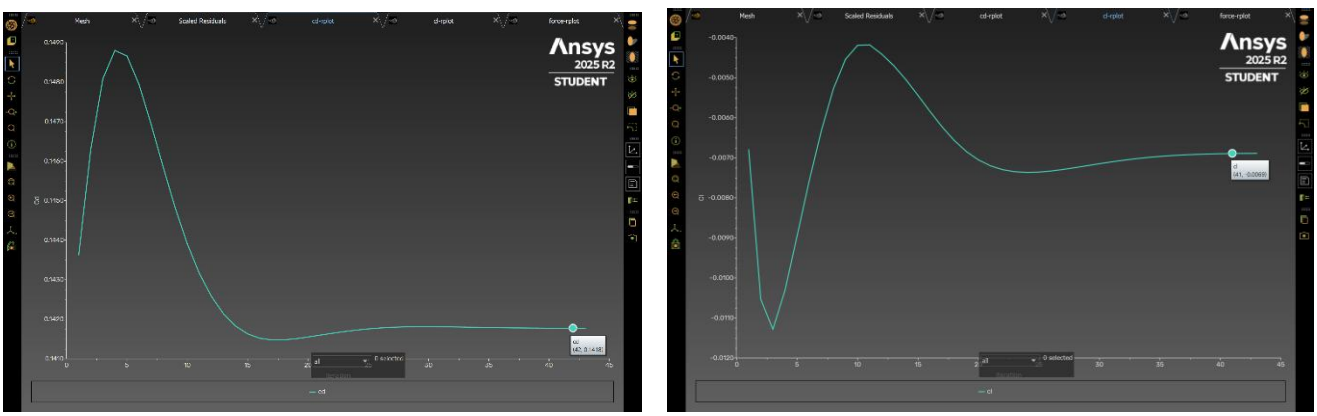


Table 9:The Cd and Cl of tail airfoil

Drag Coefficient (Cd = 0.1418):

- Much lower than your previous result of Cd = **0.1418**.
- Enhanced streamlining of the geometry
- Reduced frontal area
- Better boundary layer control or mesh refinement
- The lower drag coefficient indicates less resistance to forward motion, which improves fuel efficiency and overall flight performance.

Lift Coefficient (Cl = -0.0069):

- This value is near zero, with a slight negative sign, which may indicate:
  - A symmetric airfoil aligned with the freestream flow
  - Minor downward force due to flow asymmetry or mesh artifacts
- Under level flight conditions with a near-zero angle of attack, this Cl is expected.
- However, during takeoff, the aircraft pitches upward, increasing the angle of attack and generating a higher Cl.
- Lift is the key force that allows the aircraft to rise toward sea level — controlled by the pilot through angle adjustment.

3.3.4. The Ramjet

3.3.4.1. External of Ramjet

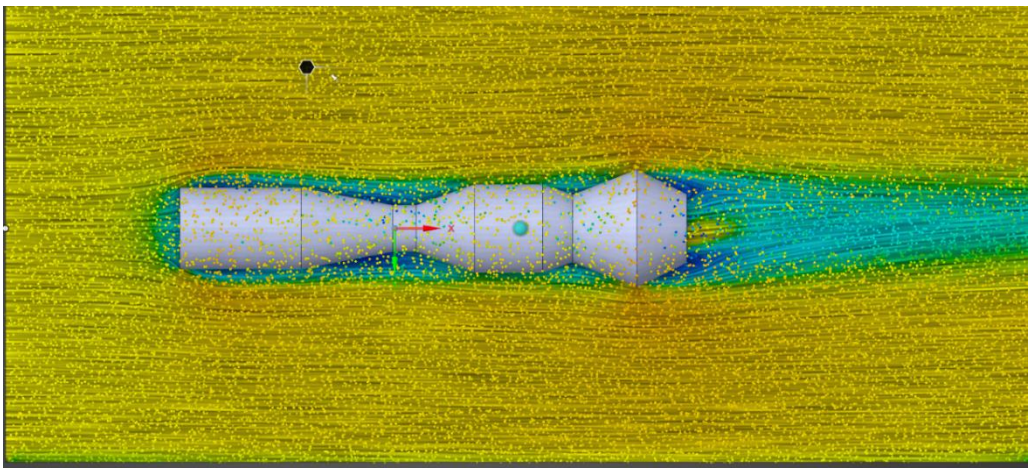
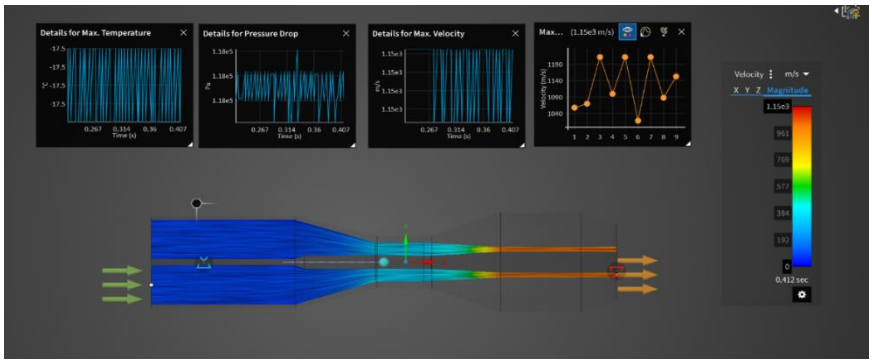


Figure 26:The external of ramjet

- **Pressure drops across the ramjet**: The outlet region shows predominantly blue tones, indicating that the static pressure at the outlet is significantly lower than at the inlet. This pressure reduction is consistent with accelerated flow and expansion through the ramjet.
- **Smooth airflow**: The streamlines and particle traces around the ramjet are continuous and well-aligned, suggesting minimal turbulence and efficient flow shaping. The absence of separation or recirculation zones confirms that the ramjet geometry supports stable external flow.

3.3.4.2. Internal of ramjet

- **Velocity**



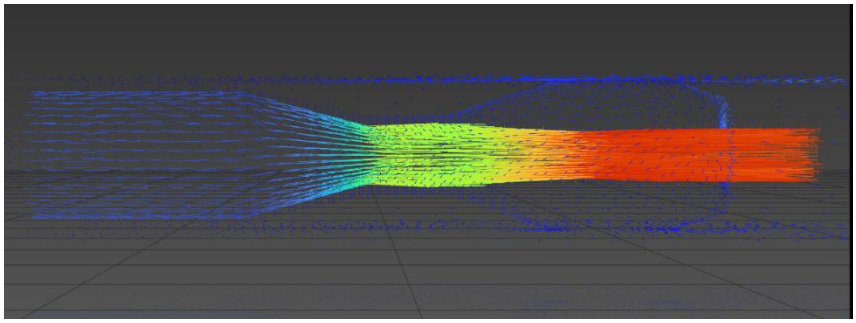


Figure 27:The speed outlet of ramjet

Flow Configuration:

- Fluid enters from the left, accelerates through a narrow constriction, and exits to the right.
- The geometry resembles a ramjet or venturi, designed to manipulate flow speed and pressure.

Velocity Visualization:

- Color gradient from blue (low velocity) to red (high velocity) shows acceleration through the throat.
- Max velocity reaches 1150 m/s, indicating either compressible flow or high inlet energy.
- Vector field confirms directional acceleration and smooth flow alignment with geometry.
- For ramjet of supersonic, we have 2 section the first section from subsonic to sonic flow and the second section from sonic to supersonic. To M1=0.1974 (Blue flow) to M2=1 (Green flow) to M3=2.6374 (Red flow).

Geometry

- The shaft at the middle in blade, machine that has ability to change air pressure to make power that could make higher speed faster than the sound speed C =335km/h.

- **Static Pressure**

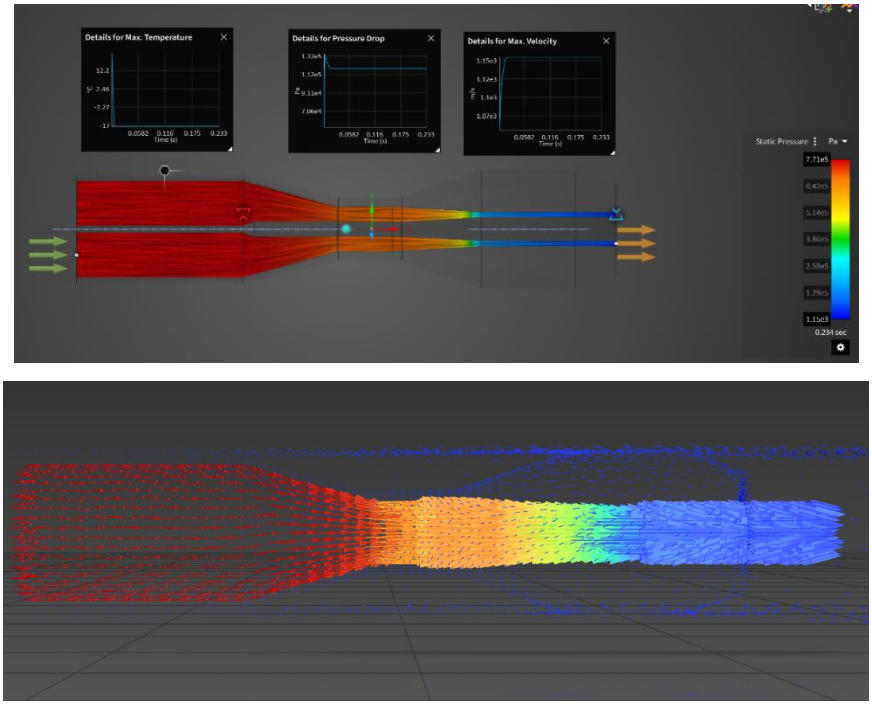


Figure 28:The static pressure outlet of ramjet

Flow Setup:

- Fluid flows through a constricted channel or ramjet, accelerating as it moves from left to right.
- The simulation visualizes static pressure, with a color gradient from red (high pressure) to blue (low pressure).
- Streamlines show how fluid particles move through the geometry.
- Red zones near the inlet and walls indicate high static pressure, where the flow slows down or compresses.
- Blue zones in the throat and outlet show low pressure, where the flow speeds up.

Bottom Visualization:

- Vector arrows show flow direction and magnitude.
- The color gradient again represents static pressure.
- Arrows bend and accelerate through the throat, confirming smooth flow and no separation

The simulation visualizes fluid flow through a constricted ramjet, highlighting static pressure distribution and velocity behavior. At the inlet, the flow exhibits high static pressure (red zones), which gradually decreases as the fluid accelerates through the narrow throat, transitioning to low-pressure regions (blue zones) at the outlet. Streamlines and vector arrows confirm smooth, attached flow with no signs of separation, indicating effective geometry and stable boundary conditions.

Graphs of temperature, pressure drop, and velocity over time show consistent convergence, with pressure drop stabilizing around 7.06e4 Pa and velocity peaking near 11.5 m/s. These results suggest efficient energy conversion from pressure to kinetic energy, validating the ramjet's aerodynamic performance.

- **Velocity, Static Pressure, and Temperature**

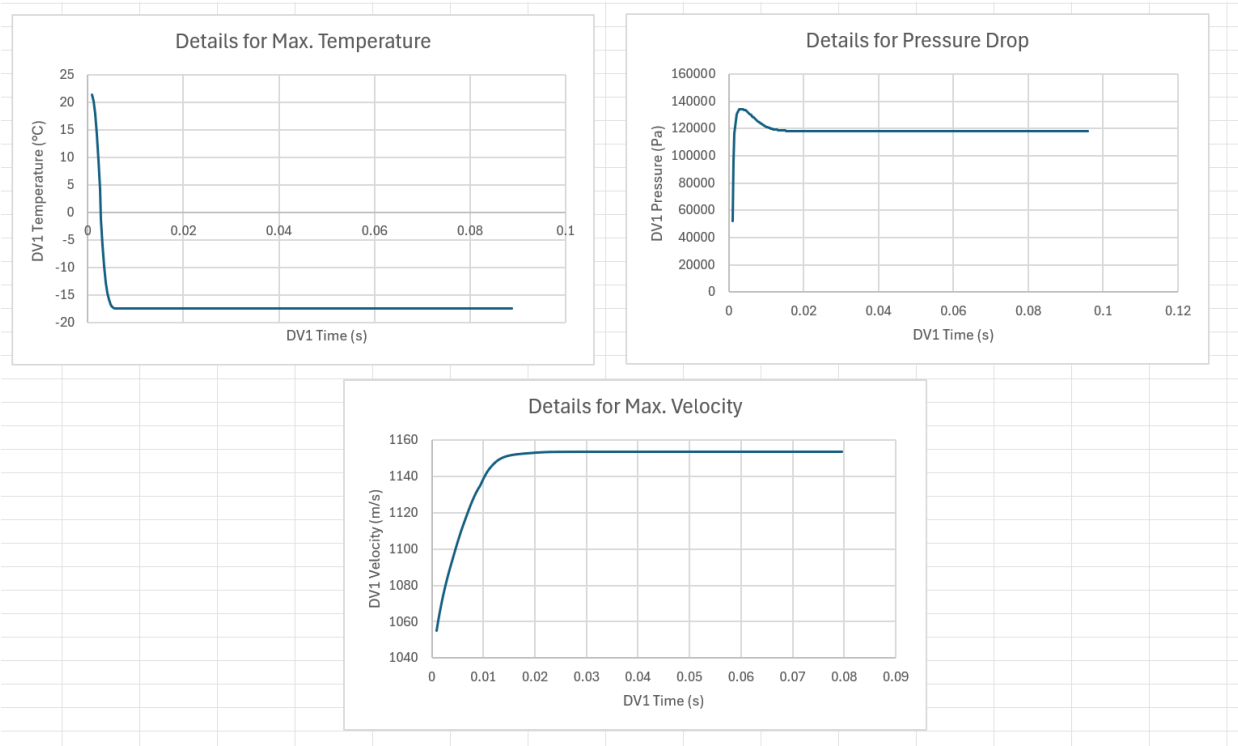


Table 10:The temperature, pressure drop, and velocity of ramjet

Velocity Analyze:

- **Initial peak velocity:** The graph titled *Details for Max. Velocity* shows a rapid increase from approximately 1040 m/s to 1150 m/s within the first 0.02 seconds. This sharp acceleration reflects the initial thrust phase, likely occurring during takeoff when the aircraft pitches upward at an angle between 5° and 15°.
- **Stabilized velocity:** After the initial surge, the velocity curve levels off around 1150 m/s, indicating the establishment of steady supersonic flow. This corresponds to cruising flight at a near-zero pitch angle, where the aircraft maintains stable speed and aerodynamic balance.

Temperature Variation:

- **Initial temperature:** The *Details for Max. Temperature* graph begins at approximately 22°C, representing ambient or baseline conditions at startup.

- **Final temperature:** Within the first 0.02 seconds, the temperature drops sharply to around −18°C, then stabilizes.
    - This cooling is caused by convective heat transfer and expansion effects in high-speed flow zones.
    - The drop is especially relevant near sea-level flight, where ambient cooling helps reduce thermal stress on the engine, Ramjet, and fuselage.

Pressure Drop Dynamics:

- **Initial pressure peak:** The *Details for Pressure Drop* graph shows a rise to approximately 140,000 Pa, followed by a gradual decrease.
- **Stabilized pressure:** The curve settles around 120,000 Pa, indicating that the system reaches flow equilibrium after the transient startup phase.
- **Interpretation:** The initial fluctuation reflects shock wave formation and rapid expansion, typical of supersonic ramjet behavior.
    - Managing these pressure changes is essential to prevent structural damage and ensure aerodynamic stability.
    - The final static pressure is relatively low compared to the inlet, confirming effective flow acceleration and energy conversion.
- **Engineering implication:** This temperature behavior is critical for thermal management, affecting material selection and structural integrity under supersonic conditions.

IV. Result

The simulation successfully modeled the flow through a ramjet nozzle transitioning from subsonic to supersonic conditions. Using MATLAB and isentropic relations, the flow properties were computed at three key sections: inlet, throat, and outlet.

Section	Mach Number	Velocity (m/s)	Pressure (kPa)	Temperature (K)
Inlet	0.1974	63.27	54.00	255.65
Throat	1.0000	320.50	23.73	213.04
Outlet	2.6374	1221.50	0.609	611.31

- **Mach progression** confirms correct acceleration: subsonic → sonic → supersonic.
- **Velocity increases** from 63.27 m/s to 1221.5 m/s, consistent with nozzle expansion.
- **Static pressure drops** from 54.0 kPa to 0.609 kPa, indicating efficient thrust generation.
- **Temperature behavior** reflects energy conversion and heat addition: lowest at throat, highest at outlet due to combustion.

By CFD of airfoil, tail airfoil, and the ramjet, we can find the **Cd (Coefficient drag)** and **Cl (Coefficient lifting)** and fit the general machine in the real life.

Type	Cd (Coefficient drag)	Cl (Coefficient lifting)
Airfoil	0.5338,	0.0426,
Tail Airfoil	0.1418	-0.0069

Airfoil

- **Cd = 0.5338:** This is relatively high for a clean wing, but acceptable for supersonic flow where wave drag and skin friction dominate.
- **Cl = 0.0426:** Very low lift, suggesting the airfoil is designed for stability and control at high speed, not for generating lift during takeoff.
-

Tail Airfoil

- **Cd = 0.1418:** Low drag, which is ideal for control surfaces.
- **Cl = –0.0069:** Slight negative lift means the tail is producing a downward force, which helps balance the aircraft's nose-up pitching moment this is typical in supersonic jets.

Graphical Analysis:

Moreover. We have pressure. Velocity, Temperature graphs which depend on time and high of sea-level. By these graphs, we are able to analysis, guess what would be happened, known about the detail of machine-like friction, failure point, and understood how to improve the plane and machine in the future.

V. Conclusion

This study successfully applied gas dynamics principles to simulate supersonic flow through a ramjet nozzle. This project had had to applied many fields into this case study like Material Science, Gas Dynamic, Mechanical Design, Structure mechanics, fluid and Thermal and another subject. Each subjects have many advantages and very importance. Moreover, this study is learnt about the basic, and know how to apply the theoretical, lesson into the real environment. It is a source, knowledge for student to research or who are interesting in the supersonic flight, pushed to the next level and more complex related to the supersonic flight.

The creation of a supersonic aircraft demands careful study of every component **airfoil, tail airfoil, fuselage structure, nozzle, and jet engine**. Each part plays a critical role in performance and must be evaluated with precision to ensure safety, efficiency, and reliability.

Achievements:

- Modeled isentropic flow from subsonic to supersonic regimes.
- Verified choking at the throat and supersonic expansion at the outlet.
- Applied area–Mach relations and energy equations to compute flow properties.
- Integrated aerodynamic theory with practical engine design.

Engineering Insight:

- The pressure and temperature profiles confirm efficient thrust generation.
- The use of titanium materials and optimized fuselage geometry supports high-speed flight.
- The F-16's engine cycle and ramjet integration were accurately represented using Brayton cycle principles.

Future Skills and Development

- Trying to improve the Structure mechanic by draw it more detail and more complex.
- Using the Ansys to simulation (Not Student version because it is not allowing student to analysis with the complex geometry.
- Studying more case and more angle of plane when the plane land off and take off land.
- Learning and put more conditions to get the actual results.
- Know how to write the MATHLAB code more because it is very useful.
- Calculate it without perfect case, by applied more evidence, environment case.

VI. References

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VII. Appendix

Cd and Cl:

$$U(y) = U_{inf} \times [0.75 - 0.25 * cos\left(pi * \frac{y}{H}\right)]$$

$$Cd = \frac{D}{\left(0.5 * rho * U_{inf}^2 * C\right)} \quad \text{and} \quad D = \rho \cdot \int_0^H (U\infty - U(y)) \cdot U(y)\, dy$$

$$Cl = \frac{L}{\left(0.5 * rho * U_{inf}^2 * C\right)} \quad and \quad L = \rho \cdot \int_0^H V(y) \cdot U(y)\, dy$$

By using Momentum Equation:

$$\sum F = \frac{d}{dt}\int Rho * VdV + \int Rho * V * (V.n)dA$$

Isentropic Flow:

$$\frac{T_{01}}{T_1} = \left(1 + \frac{\gamma - 1}{2}\right) \times (M_1)^2$$

$$\frac{P_{01}}{P_1} = \left(\frac{T_{01}}{T_1}\right)^{\gamma/\,\gamma-1}$$

$$\frac{P_2}{P_1} = \frac{1 + (\gamma \times M_1^2)}{1 + \gamma}$$

Relation of area and Mach number:

$$\frac{A_3}{A^*} = \frac{1}{M}\left[\frac{2}{\gamma + 1} \times 1 + (\gamma - 1)\frac{M^2}{2}\right]^{\frac{\gamma+1}{2(\gamma-1)}}$$

Gas Dynamic:

$$T_1 = T_0 - L \times h$$

$$P_s = P_0\left(\frac{T_1}{T_0}\right)^{\frac{\gamma}{\gamma-1}}$$

$$C = \sqrt{\left(\gamma \times R_{gas} \times T_1\right)}$$

$$V = \sqrt{\left(2\frac{P_s - P_t}{\phi_1}\right)}$$

$$M = \frac{V}{C}$$

Where:

- T0: Reference temperature at sea level [K]
- L: Temperature lapse rate [K/m]
- h: Altitude above sea level [m]
- Ps: Static Pressure [Pa]
- Pt: Total Pressure [Pa]
- φ: Density function [kg/m3]
- V1: Air speed [m/s2]
- M1: Mach Number
- T1: Temperature inlet [K]
- P1: Pressure inlet [Pa]
- C: Sound speed [m/s2]
- T01, P01: Total (stagnation) temperature and pressure at station 1 [K]; [Pa]
- T2, P2: Static temperature and pressure at station 2 [K]; [Pa]
- γ: Ratio of specific heats (Gamma) $that\ \gamma = 1.4\ (\ Air)$
- d is Integrate momentum deficit
- L is Integrate momentum lifting
- U_{inf}: Speed inlet of air
- Rho: the air density
- ∑F: Sum of external $that\ \sum F = F_B + F_S$
- CV: Control volume (region of fluid we're analyzing)
- CS: Control surface (boundary of the control volume)
- V: Velocity vector
- n: Outward normal vector on the surface
- g: Gravity $that\ g = 9.81 kg/s$^2